

Remote-tracking branches

Your laptop keeps local branches like main

Fetch, track, compare

- Fetch downloads new commits and updates origin slash star refs only
- Pull is fetch plus integrate into your current branch, usually a merge
- Set upstream with push minus u or branch minus u so status and push know which remote
- Prune stale origin refs after branches are deleted on the host

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Remote-tracking branches

See `main` vs `origin/main` vs the real remote — practice `fetch` vs `pull` and read ahead/behind.

Starter example: Local `main`, tracking `origin/main`, and the real remote all match at `c300`. Commit locally (ahead), or simulate a teammate push (behind), then `fetch/pull/push`.

Load starter example

Challenges 0 / 22 cleared

Quiz: `origin/main`: `origin/main` is a? Answer: remote-tracking branch

Answer

Show hintCheckNextIdle

Quiz: `origin/main`Quiz: `fetch`Quiz: `pull`Starter syncedGo aheadFetch keeps localBehind after teammatePush publishesPush rejectedPull FFPull mergeQuiz: aheadQuiz: behindQuiz: upstreamFetch ≠ pullQuiz: not localSynced after pushQuiz: pull partsStale trackingQuiz: `-u`Three equalRoundtrip

Three places

fetch

Updates `origin/*` only — does not move your local branch.

Synced starterLocal aheadBehind (teammate)Diverged

synced with `origin/main`

pull

Fetch + integrate into current branch (merge by default in this lab).

Remote-tracking lab starter

learn_git — remote tracking

Real Git practice

```
wsl - module16-remote-tracking/examples/remote_branch

# Track A - real Unix
# real shell session (Track A)
# practice sandbox: /tmp/git-rt-l2gewC

$ git init

$ echo "v1" > README.md

$ git add README.md

$ git commit -m "init"

$ git branch -M main

$ git init --bare ../origin.git

$ git remote add origin ../origin.git

$ git push -u origin main

$ git clone ../origin.git ../teammate

$ echo "v2 remote" >> README.md

$ git add README.md

$ git commit -m "remote advance"

$ git push origin main

$ git fetch origin

$ git branch -r
origin/main

$ echo "local work" >> README.md

$ git add README.md

$ git commit -m "local commit"

$ git branch -vv
* main 359412f [origin/main: ahead 1, behind 1] local commit

$ git checkout -b feature-tmp

$ git push -u origin feature-tmp

$ git push origin --delete feature-tmp

$ git fetch origin

$ git remote prune origin

$ git branch -r
origin/main
```

Real shell — fetch, branch refs, prune

Real Git practice — try these

```
# git init — create a new repository here  
git init
```

```
# echo ... > README.md — first commit on main  
echo "v1" > README.md
```

```
git add README.md
```

```
git commit -m "init"
```

```
# git branch -M main — name the default branch main  
git branch -M main
```

```
# git init --bare ../origin.git — bare remote (simulates GitHub)  
git init --bare ../origin.git
```

Pitfalls to watch

- Fetch is safe
- Do not confuse origin slash main with your local main
- Prune only removes stale remote-tracking refs, not your local branches
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and clear a few ahead-or-behind challenges
- On real Git, run fetch, read branch minus v v, and prune a deleted remote branch
- When you are ready